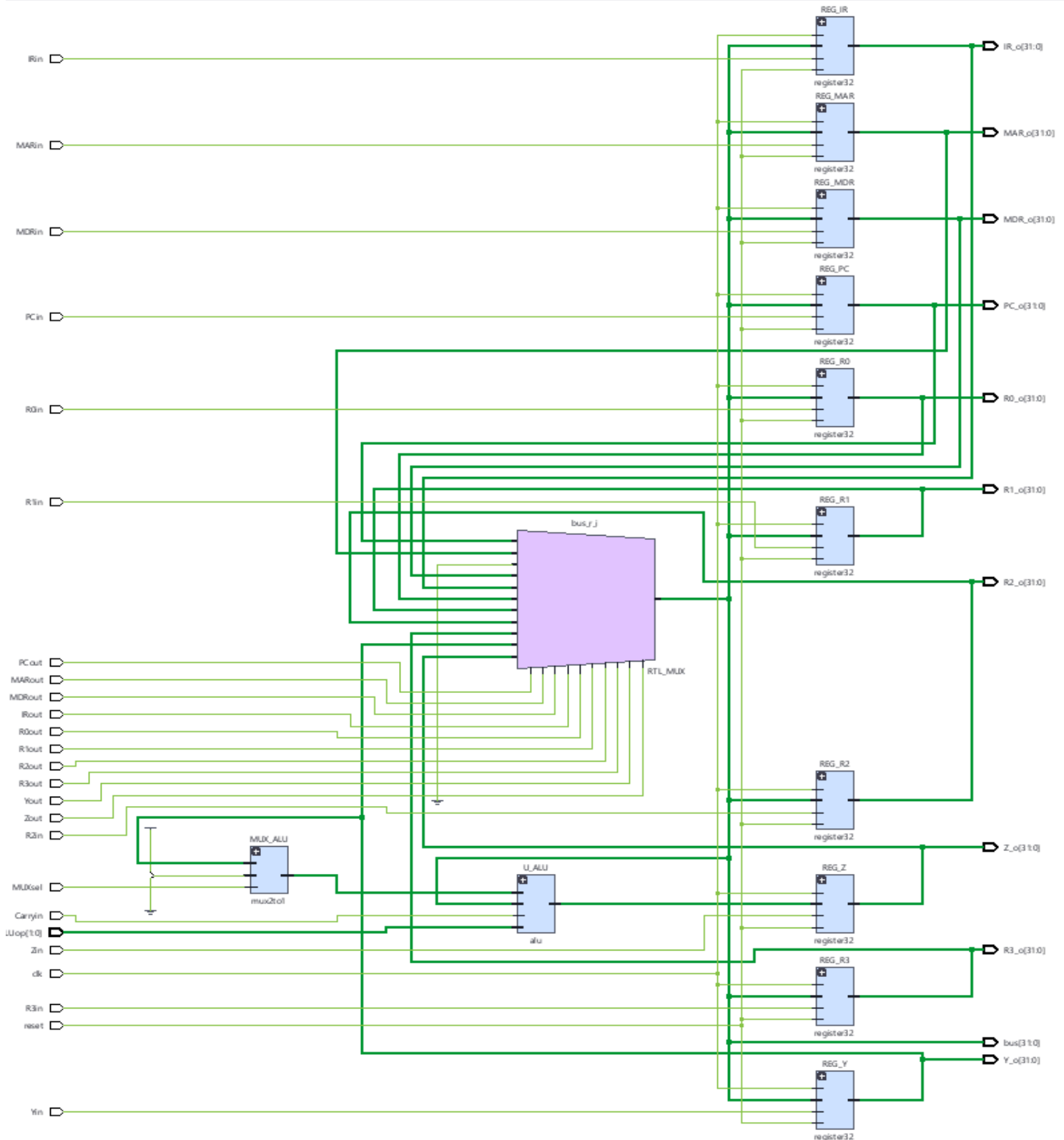


Computer Architecture Assignment 2 :

Single-Bus Datapath

- Vivado Simulation Results

Name: Vaghasiya Pal (B24CS1078)





```
===== INITIAL VALUES =====
R1 = 10   R2 = 3   PC = 100

----- (i) R2 <- R1 -----
R1=10 R2=10 (expected R2=10)

----- (ii) R3 <- R1 + R2 -----
cycle1: Y=10 (expected 10)
cycle2: Z=13 (expected 13)
cycle3: R3=13 (expected 13)

----- (iii) R3 <- R1 - R2 -----
cycle1: Y=10 (expected 10)
cycle2: Z=7 (expected 7)
cycle3: R3=7 (expected 7)

----- (iv) PC <- PC + 4 -----
cycle1: Y=100 (expected 100)
cycle2: Z=104 (expected 104)
cycle3: PC=104 (expected 104)

===== FINAL VALUES =====
R1=10 R2=3 R3=7 PC=104
```